Write your name here

Surname	Other names

Pearson Edexcel

Level 1/Level 2 GCSE (9-1)

Mathematics

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Hannah Kettle MATHS TUTORIALS

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Higher Tier

Half Past Paper – Week 6

Total Marks

Q	Topic		Sparx Code
2	Expanding single brackets	2	U179
4	Tree diagrams for independent events	2	U558
6a	Probabilities of mutually exclusive events	4	U683
6b	Writing probabilities as fractions, decimals and percentages	1	U510
8	Angles in polygons	5	U427
10	Finding the lowest common multiple	3	U751
12a	Finding equations of linear real-life graphs	2	U862
12b	Finding equations of linear real-life graphs	1	U862
12c	Finding equations of linear real-life graphs	1	U862
14	Using the product rule for counting	2	U369
16a	Quadratic sequences, Simultaneous equations involving quadratics	4	U206,U547
16b	Position-to-term rules for quadratic sequences	2	U206
18a	Substituting, Finding approximate solutions to equations using iteration	2	U585,U168
18b	Finding approximate solutions to equations using iteration	1	U168
18c	Finding approximate solutions to equations using iteration	3	U168
20	Venn diagrams	5	U476
	Total	40	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

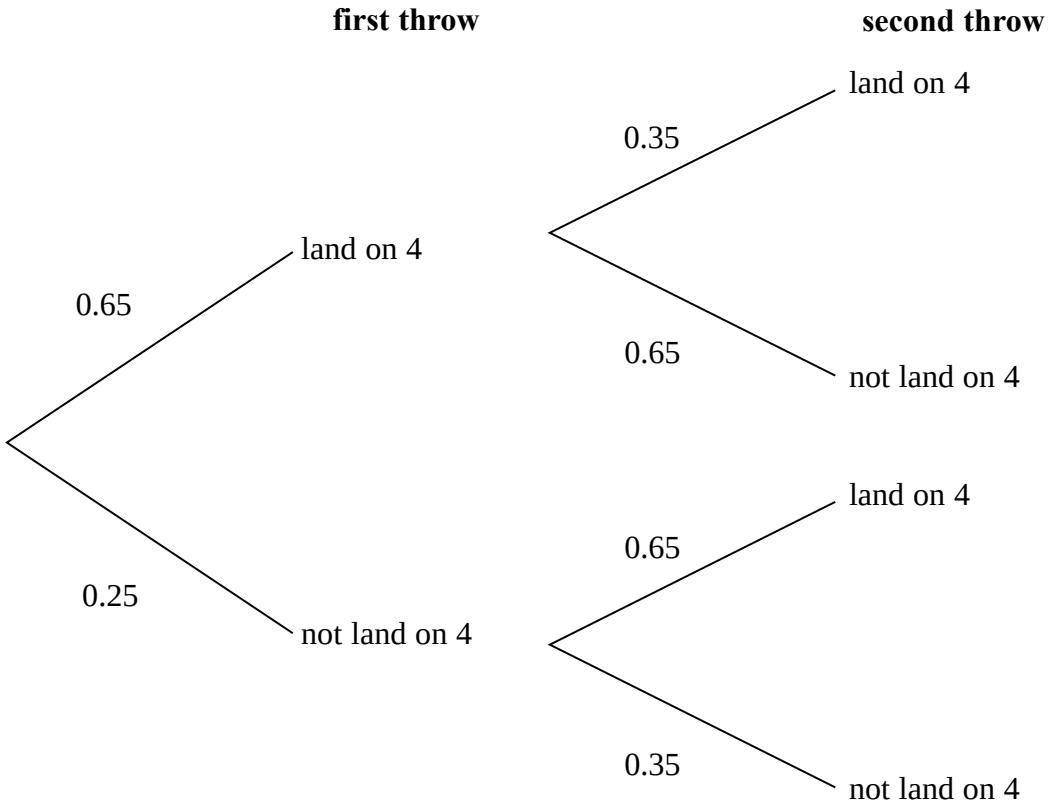
2 Expand and simplify $5(p + 3) - 2(1 - 2p)$

.....

(**Total for Question 2 is 2 marks**)

4 When a biased 6-sided dice is thrown once, the probability that it will land on 4 is 0.65
The biased dice is thrown twice.

Amir draws this probability tree diagram.
The diagram is **not** correct.



Write down **two** things that are wrong with the probability tree diagram.

1 .....

.....

2.....

.....

(**Total for Question 4 is 2 marks**)

6 There are some counters in a bag.
The counters are red or white or blue or yellow.

Bob is going to take at random a counter from the bag.

The table shows each of the probabilities that the counter will be blue or will be yellow.

Colour	red	white	blue	yellow
Probability			0.45	0.25

There are 18 blue counters in the bag.

The probability that the counter Bob takes will be red is twice the probability that the counter will be white.

(*a*) Work out the number of red counters in the bag.

.....

(**4**)

A marble is going to be taken at random from a box of marbles.
The probability that the marble will be silver is 0.5.

There must be an even number of marbles in the box.

(*b*) Explain why.

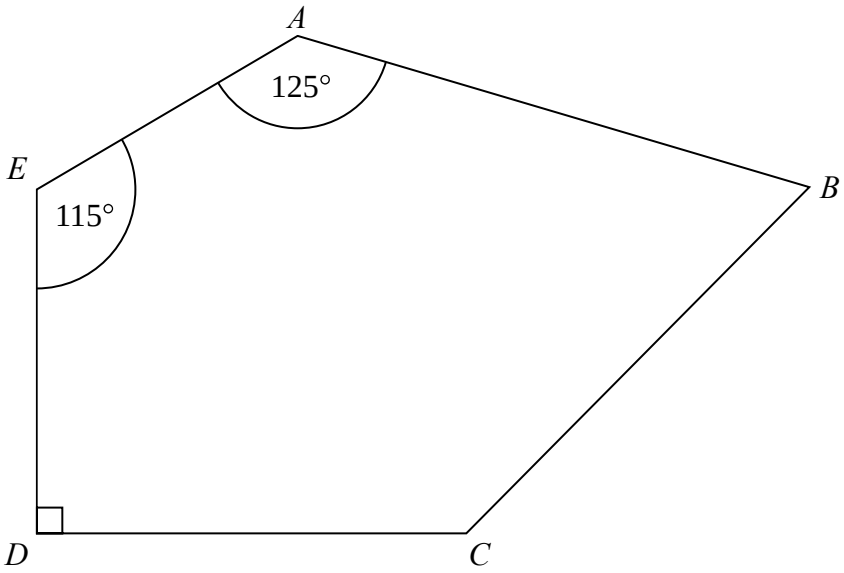
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(**1**)

(**Total for Question 6 is 5 marks**)

8 *ABCDE* is a pentagon.



Angle *BCD* = 2 × angle *ABC*

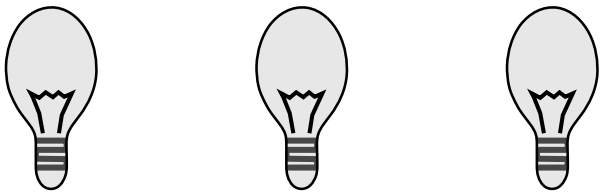
Work out the size of angle *BCD*.
You must show all your working.

..... °

(**Total for Question 8 is 5 marks**)

10 Here are three lamps.

lamp **A** lamp **B** lamp **C**



Lamp **A** flashes every 20 seconds.
Lamp **B** flashes every 45 seconds.
Lamp **C** flashes every 120 seconds.

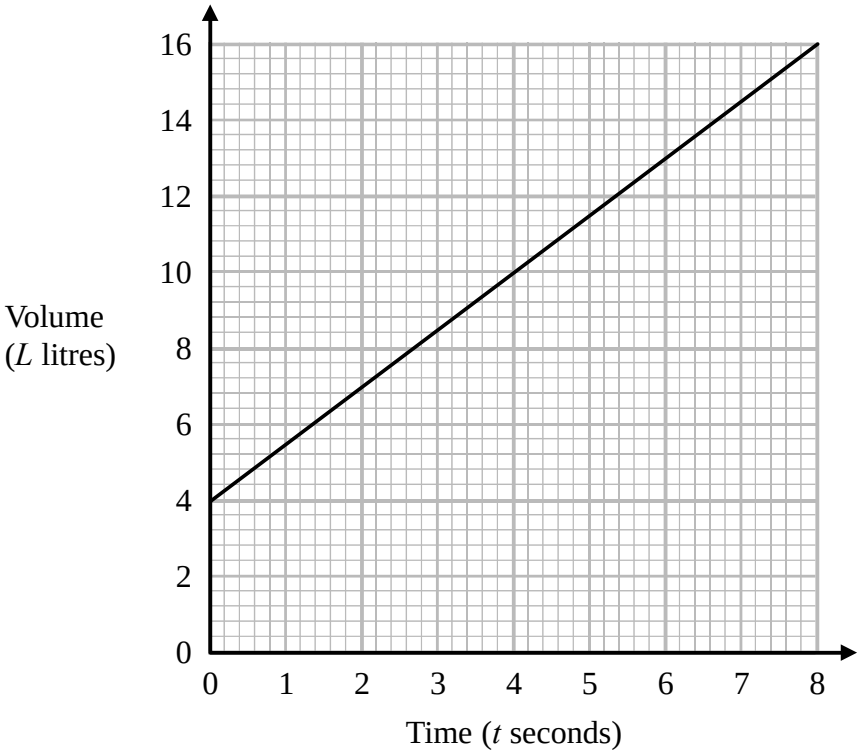
The three lamps start flashing at the same time.

How many times in one hour will the three lamps flash at the same time?

.....

(**Total for Question 10 is 3 marks**)

12 The graph shows the volume of liquid (L litres) in a container at time t seconds.



(a) Find the gradient of the graph.

.....

(**2**)

(b) Explain what this gradient represents.

.....

.....

(**1**)

The graph intersects the volume axis at $L = 4$

(c) Explain what this intercept represents.

.....

.....

(**1**)

(**Total for Question 12 is 4 marks**)

14 There are 16 hockey teams in a league.
Each team played two matches against each of the other teams.

Work out the total number of matches played.

.....

(**Total for Question 14 is 2 marks**)

16 The nth term of a sequence is given by $an^2 + bn$ where a and b are integers.

The 2nd term of the sequence is –2.
The 4th term of the sequence is 12.

(a) Find the 6th term of the sequence.

.....

(**4**)

Here are the first five terms of a different quadratic sequence.

0 2 6 12 20

(b) Find an expression, in terms of n, for the nth term of this sequence.

.....

(**2**)

(**Total for Question 16 is 6 marks**)

18 (*a*) Show that the equation $x^3 + x = 7$ has a solution between 1 and 2.

(2)

(*b*) Show that the equation $x^3 + x = 7$ can be rearranged to give $x = \sqrt[3]{7 - x}$

(1)

(*c*) Starting with $x_0 = 2$,
use the iteration formula $x_{n+1} = \sqrt[3]{7 - x_n}$ three times to find an estimate for a solution of $x^3 + x = 7$

.....

(3)

(Total for Question 18 is 6 marks)

20 50 people were asked if they speak French or German or Spanish.

Of these people,

31 speak French
2 speak French, German and Spanish
4 speak French and Spanish but not German
7 speak German and Spanish
8 do not speak any of the languages
all 10 people who speak German speak at least one other language

Two of the 50 people are chosen at random.

Work out the probability that they both only speak Spanish.

.....

(**Total for Question 20 is 5 marks**)

TOTAL FOR PAPER IS 40 MARKS